

Project - LTI identification

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I. SYSTEM DESCRIPTION

The device studied throughout this project is a linear system with a static nonlinear feedback loop. It has a single input and a single output and the feedback is known to follow the input-output relation given by

$$y(t) = u^3(t). \quad (1)$$

The objective will be to identify the linear part of the system and to have a measure of the nonlinearity and noise levels. The knowledge of (1) will not be used, except for a small parenthesis about odd and even nonlinearities.

The measurement device (referred to as the *PXI setup*) allows to send only multisines and their RMS value cannot exceed 1 V. Its ADCs have a range that can be chosen and can sample at high frequencies (> 1 GHz).

This report deliberately focusses on the theoretical points and the results rather than the implementation because the Matlab scripts and the measured data are available on GitHub¹.

II. EXCITATION SIGNAL

Before making any useful measurement, the parameters of the excitation signal must be chosen to match to the DUT parameters. The rms of the applied signal must indeed be determined to use the full range of the ADC while avoiding saturation of the system. The poles of the system must also be roughly known to determine the band of interest, making sure that the system's dynamics are measured.

A. Band of Interest

To determine the sampling frequency, a multisine excited up to 40 kHz has been applied to the DUT. Both input and output of the system are shown in the frequency domain in Fig. 1.

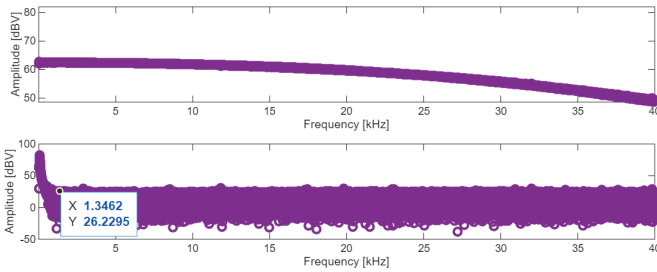


Fig. 1. Excitation spectrum (top) and output spectrum (bottom).

The measured input spectrum is not flat because of the ZOH: the continuous signal is discretized by the *PXI setup* and the

hold operation is equivalent with a sinc filtering in frequency. This effect could be reduced with a higher sampling frequency.

The output spectrum shows that the system's dynamics happen at frequencies below ~ 1.25 kHz. The excitation signals used will therefore stop at this frequency. To mitigate the effect of the ZOH, the sampling frequency is set to 5 kHz.

B. Amplitude

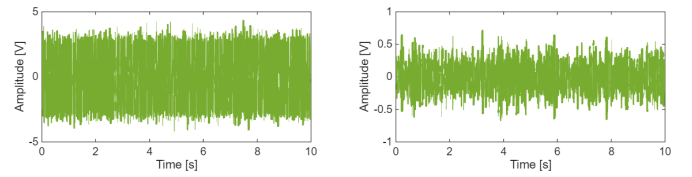


Fig. 2. Excitation signal (left) and output signal (right).

The same signal as the one measured for Sec. II-A but is shown in the time domain in Fig. 2. It can be seen that the input signal do not go higher than 5 V and the output voltage does not cross 1 V. This allowed to set the range of the ADCs of the *PXI setup*.

C. Transient Removal

Because only multisines can be applied, every measurement is done for multiple periods and the first ones are systematically removed until the transient is removed. To check this, the difference between each period is plotted, as shown in Fig. 3 and periods are removed until the shown plot is reduced to a flat curve.

III. NONLINEAR DISTORTION AND NOISE ANALYSIS

Two methods are used to distinguish the nonlinear distortions from the noise. The *fast method* uses a single experiment and allows to split the output spectrum in linear, nonlinear and noise contributions. The *robust method* uses multiple experiments and computes an estimate of the variances of nonlinear distortion and noise together with an nonparametric estimate of the frequency response function (FRF).

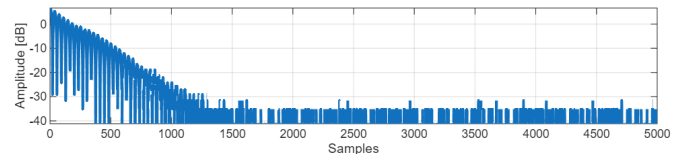


Fig. 3. Difference between 1st and 2nd periods of the output signal in time domain.

¹<https://github.com/e-colot/LTI-identification>

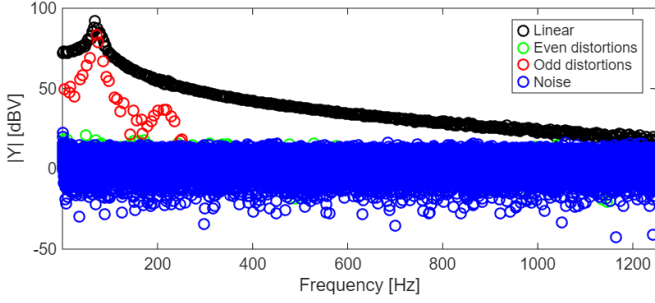


Fig. 4. Results of the fast method with the feedback given by (1).

A. Fast Method

The excitation signal is an odd multisine with holes. This means that only odd frequencies are excited and some of those odd frequencies are also unexcited in order to have a measure for odd nonlinear distortions. During the excitation generation, 1 odd bin out of 4 is randomly chosen to be not excited.

The FFT of multiple periods are taken, effectively dividing the spectral resolution as $\Delta_f = f_s/N$ with a larger N . The added bins are not linear combinations of input frequencies and therefore only contain noise, under the assumption that the DUT is a *PISPO*² system.

The spectrum of the output signal is then divided into 4 categories:

- **Linear**, for bins that were excited.
- **Even distortions**, for bins that are a combination of an even number of excited bins.
- **Odd distortions**, for bins that are a combination of an odd number of excited bins.
- **Noise**, for bins that are added by the multiple periods.

Fig. 4 shows that even distortion is close to the noise level and therefore does not impact much the signal. Odd distortion on the other hand is quite high and is close to the level of the linear part near its peak.

This behavior is explained by (1): because the third power of the output is fed back to the input, it contains sinusoids at the sum of 3 frequencies of the output. As the output only contains odd frequencies, it results in another odd frequency and the main nonlinearities therefore appears at odd bins.

To validate this explanation, the same experiment has been applied to the system with this time as feedback

$$y(t) = u^2(t), \quad (2)$$

which gave the results shown in Fig. 5.

The feedback now excites even frequencies and therefore has more even distortions than the one of Fig. 4.

B. Robust Method

This method relies on computing a FRF for each of the P periods and the M realizations. The variance across the periods for realization m is called $\hat{\sigma}_n^{2[m]}$ and is only due to noise. This is because the nonlinear contributions are

²period in, same period out

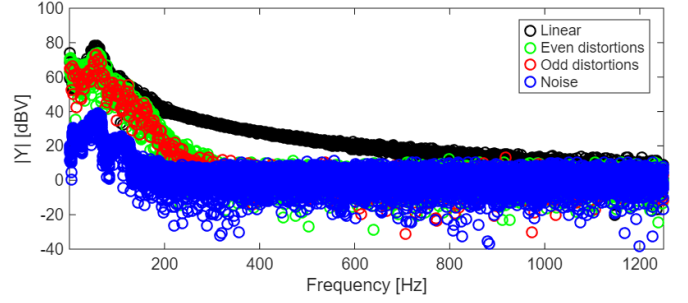


Fig. 5. Results of the fast method with the feedback given by (2).

deterministic and depend only on the input signal, which is the same for the whole realization.

$$\hat{G}^{[m,p]} = \frac{Y^{[m,p]}}{U^{[m]}} \quad (3)$$

$$\hat{G}^{[m]} = \frac{1}{P} \sum_{p=1}^P \hat{G}^{[m,p]} \quad (4)$$

$$\hat{\sigma}_n^{2[m]} = \frac{1}{P(P-1)} \sum_{p=1}^P \left| \hat{G}^{[m,p]} - \hat{G}^{[m]} \right|^2 \quad (5)$$

These noise variances on a single realization $\hat{\sigma}_n^{2[m]}$ are then averaged across the realizations and divided by M because the averaging reduces the variance. This gives the total noise variance σ_n^2 .

$$\hat{\sigma}_n^2 = \frac{1}{M^2} \sum_{m=1}^M \hat{\sigma}_n^{2[m]} \quad (6)$$

The averaged FRF of a single realization $\hat{G}^{[m]}$ does not contain noise (at least in expected value). But between realizations, they have different nonlinear effects as the phases of the excitation signals have changed. The variance between those therefore is due to nonlinear distortion and averaging it out gives a FRF estimate that is noiseless and purely linear, if M and P are large enough to be close to the expected value.

$$\hat{G}_{ML} = \frac{1}{M} \sum_{m=1}^M \hat{G}^{[m]} \quad (7)$$

$$\hat{\sigma}_{ML} = \frac{1}{M(M-1)} \sum_{m=1}^M \left| \hat{G}^{[m]} - \hat{G}_{ML} \right|^2 \quad (8)$$

A full multisine (excited up to 1.25 kHz as explained in Sec. II-A) has been applied to the DUT for $P = 20$ periods and $M = 20$ realizations. The results are shown on Fig. 6.

IV. BEST LINEAR APPROXIMATION

A linearized model of the system can be determined to approximate the DUT as an LTI. The best linear approximation, or BLA, is therefore computed as the "best fit". Because nonlinear distortions are dependent on the input signal, the BLA computed here is valid for excitation signals with a similar *Riemann equivalent spectrum*.

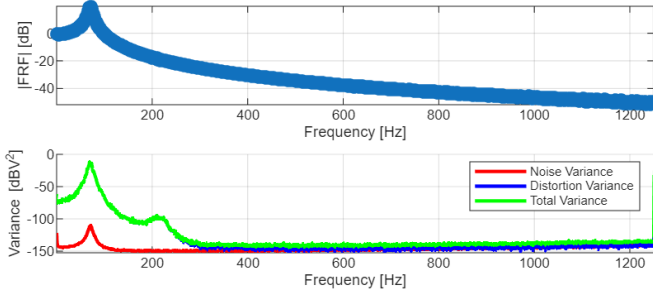


Fig. 6. Results of the robust method, with the FRF estimate (top) and the estimated variances (bottom).

A. Nonparametric Model

The result of the robust method shown in Fig. 6 is, in fact, an estimated BLA and it already comes with its uncertainty. Because of the successive averaging, it should be able to remove noise and nonlinear distortions which are zero-mean³.

B. Parametric Model

Based on the nonparametric model of Sec. IV-A, a parametric model can be derived by making a choice for the model. This will be fixed by the order of the numerator and denominator of the transfer function, respectively n_b and n_a giving the following expression

$$G(s) = \frac{\sum_{k=0}^{n_b} b_k s^k}{\sum_{k=0}^{n_a} a_k s^k}. \quad (9)$$

Different estimators are used but first, a few tricks are used for each of them and those are explained here.

a) Normalization by rms scaling: Each estimator has a regressor matrix (H or J), whose columns often have different orders of magnitude. This results in a poor conditioning and gives numerical errors. To resolve this, a change of coordinates is applied with the scaling matrix S , which scales the columns. The original coordinates H (or J) and θ are replaced by $\tilde{H}$ (or $\tilde{J}$) and $\tilde{\theta}$.

$$y = H \times \theta \quad (10)$$

$$\Rightarrow y = \tilde{H} \times \tilde{\theta} \quad \text{with} \quad \tilde{H} = H \times S \quad (11)$$

$$\Leftrightarrow \tilde{\theta} = S^{-1} \times \theta \quad (12)$$

$$\Leftrightarrow \theta = S \times \tilde{\theta} \quad (13)$$

S has been chosen to be a diagonal matrix with as elements the inverse of the rms of the columns of H , such that it is invertible

$$S = \text{diag}\left(\frac{1}{\text{rms}(H_{:,1})}, \dots, \frac{1}{\text{rms}(H_{:,n})}\right). \quad (14)$$

³It is only true in expected value so it can only be approached using a discrete number of periods and realizations

b) Use of the nonparametric estimate: All the estimators are relying on approaching the following, which arises from the definition of a transfer function. The key idea is to notice that because

$$Y(k) = G(j\omega_k) \times U(k), \quad (15)$$

a clever choice of U simplifies the choice of Y , as

$$U(k) = 1 \quad \forall k \Rightarrow Y(k) = G(j\omega_k). \quad (16)$$

c) Forcing real valued parameters: A real system is described by real coefficients so the imaginary part of θ must be equal to 0. The regressor must therefore be changed.

$$Y = H \times \theta \quad (17)$$

$$\Rightarrow \Re\{Y\} + j\Im\{Y\} = \left(\Re\{H\} + j\Im\{H\} \right) \times \left(\Re\{\theta\} + j\Im\{\theta\} \right) \quad (18)$$

$$\Leftrightarrow \begin{bmatrix} \Re\{Y\} \\ \Im\{Y\} \end{bmatrix} = \begin{bmatrix} \Re\{H\} & -\Im\{H\} \\ \Im\{H\} & \Re\{H\} \end{bmatrix} \times \begin{bmatrix} \Re\{\theta\} \\ \Im\{\theta\} \end{bmatrix} \quad (19)$$

And because $\Im\{\theta\} = 0$, it becomes

$$\begin{bmatrix} \Re\{Y\} \\ \Im\{Y\} \end{bmatrix} = \begin{bmatrix} \Re\{H\} \\ \Im\{H\} \end{bmatrix} \times \Re\{\theta\}. \quad (20)$$

Thus y and H are replaced by a concatenation of their real and imaginary parts called Y_{re} and H_{re} as follows

$$Y_{\text{re}} = H_{\text{re}} \times \Re\{\theta\}. \quad (21)$$

As explained later on, the GTLS estimator needs a covariance matrix C_J that depends on J . Because J has become J_{re} , $C_{J_{\text{re}}}$ must be derived. starting from

$$\Delta J_{\text{re}} = \begin{bmatrix} \Re\{\Delta J\} \\ \Im\{\Delta J\} \end{bmatrix} \quad \text{and} \quad C_J = \mathbb{E}\left\{ \Delta J^H \Delta J \right\}, \quad (22)$$

The covariance matrix for a real parameters estimator is

$$\begin{aligned} C_{J_{\text{re}}} &= \mathbb{E}\left\{ \begin{bmatrix} \Re\{\Delta J\}^T & \Im\{\Delta J\}^T \end{bmatrix} \begin{bmatrix} \Re\{\Delta J\} \\ \Im\{\Delta J\} \end{bmatrix} \right\} \\ &= \mathbb{E}\left\{ \Re\{\Delta J\}^T \Re\{\Delta J\} \right\} + \mathbb{E}\left\{ \Im\{\Delta J\}^T \Im\{\Delta J\} \right\}. \end{aligned} \quad (23)$$

By splitting the real and the imaginary part of ΔJ , C_J can be computed as follows.

$$\Delta J = \Re\{\Delta J\} + j\Im\{\Delta J\} \quad (25)$$

$$\begin{aligned} \Rightarrow C_J &= \mathbb{E}\left\{ \Re\{\Delta J\}^T \Re\{\Delta J\} \right\} \\ &+ \mathbb{E}\left\{ \Im\{\Delta J\}^T \Im\{\Delta J\} \right\} + j\left(\mathbb{E}\left\{ \Re\{\Delta J\}^T \Im\{\Delta J\} \right\} \right. \\ &\quad \left. - \mathbb{E}\left\{ \Im\{\Delta J\}^T \Re\{\Delta J\} \right\} \right) \end{aligned} \quad (26)$$

Finally, by comparing (24) with (26), it is found that

$$C_{J_{\text{re}}} = \Re\{C_J\}. \quad (27)$$

1) *Linear Least Squares Estimator (LLS)*: It minimizes

$$V_{\text{LLS}} = \sum_k |E(k)|^2, \quad (28)$$

where $E(k) = AY(k) - BU(k)$. The optimal parameters θ_{LLS} is found by writing the problem as

$$Y = H\theta_{\text{LLS}}, \quad (29)$$

under the assumption that a_0 is different from 0. With all the previous tricks, it becomes

$$G_{\text{BLA}_{\text{re}}} = \tilde{H}_{\text{re}} \tilde{\theta}_{\text{LLS}} \quad (30)$$

and its solution is given by

$$\theta_{\text{LLS}} = S(H_{\text{re}} \times S)^\dagger G_{\text{BLA}_{\text{re}}} \quad (31)$$

2) *Total Least Squares Estimator (TLS)*: It minimizes the same cost function

$$V_{\text{TLS}} = \sum_k |E(k)|^2, \quad (32)$$

but there is another constraint which does not require a_0 to be different from 0, given by

$$\|\theta_{\text{TLS}}\|_2^2 = 1. \quad (33)$$

The estimated parameters are then

$$\theta_{\text{TLS}} = SV_{\tilde{J}_{\text{re}}[:, \text{end}]}, \quad (34)$$

where J is the regression matrix such that

$$E = J\theta_{\text{TLS}} \quad (35)$$

and $V_{\tilde{J}_{\text{re}}}$ is the right matrix of the singular value decomposition (svd) of $\tilde{J}_{\text{re}}$. This estimator is not consistent because the covariance of J is different from the identity matrix.

3) *Generalized Total Least Squares Estimator (GTLS)*: It allows to overcome the inconsistency of the TLS estimator by using the covariance matrix of J , namely

$$C_J = \mathbb{E}\{\Delta J^H \Delta J\}, \quad (36)$$

where ΔJ contains the noisy contributions of J . The cost function is

$$V_{\text{GTLS}} = \|J\theta_{\text{GTLS}}\|_2^2, \quad (37)$$

under the constraint that

$$\|C_J^{1/2} \theta_{\text{GTLS}}\|_2^2 = 1. \quad (38)$$

With X_{re} the right matrix of the generalized svd (gsvd) of $\tilde{J}_{\text{re}}$ and $C_{\tilde{J}_{\text{re}}}^{1/2}$,

$$\theta_{\text{GTLS}} = S(X_{\text{re}}^T)_{[:, i]}^{-1} \frac{1}{D_{\text{re}}(i, i)}. \quad (39)$$

To determine i , the middle matrices C_{re} and D_{re} of the gsvd are used:

$$i = \arg \min_i \frac{C_{\text{re}}(i, i)}{D_{\text{re}}(i, i)} \quad (40)$$

4) *Bootstrapped Total Least Squares Estimator (BTLS)*: All the previous estimators tend to overemphasize high frequency errors as the regressors (H or J) become large at higher frequencies. The BTLS estimator puts a weight inversely proportional to the standard deviation of the error at each frequency.

In practice, it is the same as a GTLS estimator where

$$\tilde{J}_{\text{re}} \mapsto W \tilde{J}_{\text{re}} \quad (41)$$

with

$$W = \text{diag}(\sigma_e^{-r}). \quad (42)$$

The parameter r is chosen in $[0; 1]$ where a value close to 0 corresponds to the GTLS estimator. It is then applied iteratively until the new cost is greater or equal to the previous one.

5) *Maximum Likelihood Estimator (ML)*: This estimator maximizes the probability that the parameters θ_{ML} generated the measured data, and is therefore mathematically the best. It is however nonlinear and must be solved iteratively, just as for the BTLS. Its cost function is

$$V_{\text{ML}} = \frac{1}{2} \sum_k \frac{|e(j\omega_k, \theta_{\text{ML}})|^2}{\sigma_e^2(j\omega_k, \theta_{\text{ML}})} \quad (43)$$

and the parameter update $\Delta\theta$ is given by

$$\Delta\theta = -J^\dagger \epsilon, \quad (44)$$

where J is the jacobian of ϵ , which is itself defined as

$$\epsilon = \frac{e}{\sigma_e}. \quad (45)$$

No rms normalization is applied because there is no numerical issue and enforcing a real valued θ_{ML} is done by splitting the real and imaginary parts of J and ϵ in (44).

$$J_{\text{re}} = \begin{bmatrix} \Re J \\ \Im J \end{bmatrix} \quad \text{and} \quad \epsilon_{\text{re}} = \begin{bmatrix} \Re \epsilon \\ \Im \epsilon \end{bmatrix} \quad (46)$$

V. MODEL ORDER SELECTION

The estimators described in Sec. IV-B require a model order (n_a, n_b) to determine the right parameters $\theta = [a_0, \dots, b_{n_b}]$ of (9).

The AIC criterion is therefore used. It depends on the cost of the estimator V_{est} and the order of the model, giving an *AIC cost*, given by

$$V_{\text{AIC}} = V_{\text{est}} \left(1 + \frac{n_a + n_b}{N_e} \right). \quad (47)$$

This is computed for various values of n_a and n_b and the optimal order is the one with the smallest *AIC cost*.

The values of the order are restricted to transfer functions with more poles than zeros ($\Rightarrow n_a \geq n_b$) to have a stable system.

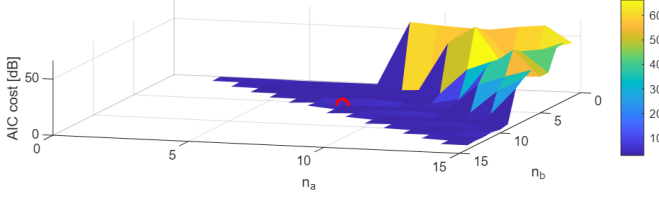


Fig. 7. AIC cost depending on the order of the transfer function for the TLS estimator. The optimal order is circled in red.

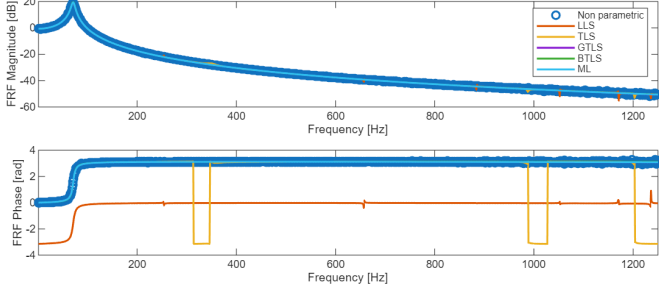


Fig. 8. Superposition of the estimated parametric FRFs amplitude (top) and phase (bottom), together with the nonparametric estimate.

VI. RESULTS

The AIC criterion described in Sec. V has been applied⁴ to every estimator of Sec. IV-B and it gives for each of them a 3D plot like the one shown in Fig. 7.

The resulting FRFs are plotted in Fig. 8. It can be seen that the LLS and TLS estimators have *bumps* where poles and zeros nearly cancel each other's. The three last estimators (GTLS, BTLS and ML) superpose perfectly, making them undistinguishable on the graph.

Fig. 9 shows the optimal order that minimizes the AIC cost (47). A few observations can be made:

- The LLS and TLS estimators both have a high number of parameters and lay close to the limit $n_a = n_b$. Fig. 10 confirms the hypothesis of imperfect pole-zero cancellations emitted based on the bumps of the FRF.
- GTLS and BTLS have a more intermediate number of parameters.
- The ML estimator uses a really low number of poles and zeros, yet achieve a performance similar to the GTLS on Fig. 8, which was expected.

Finally, the poles and zeros of the GTLS, BTLS and ML estimators are shown in Fig. 11. It is here clearly visible that

⁴for n_a and n_b going up to 15.

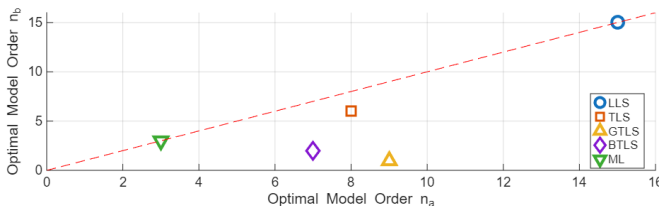


Fig. 9. Optimal orders for each estimator, according to the AIC criterion. The red dashed line is the stability limit $n_a = n_b$

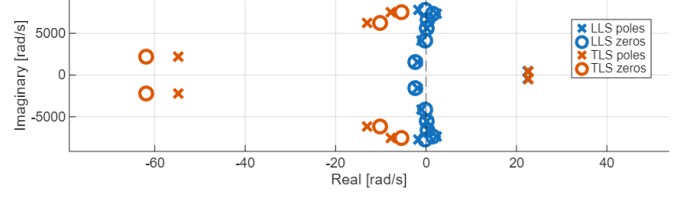


Fig. 10. Poles and zeros of the LLS and TLS estimators

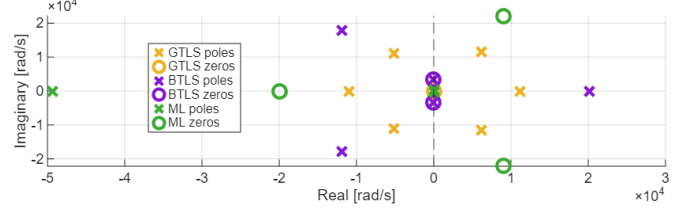


Fig. 11. Poles and zeros of the GTLS, BTLS and ML estimators

the GTLS estimator has poles in the right half-plane at high frequency, meaning the resulting system is unstable. Every estimator has poles with a positive real part but those of the ML and BTLS estimators are around 20 rad/s, which is by far *less unstable* than the one above 1000 rad/s of the GTLS.

VII. DEEPER ANALYSIS OF THE POLES

A. Poles position uncertainty

The covariance of the poles (and zeros) of the ML estimator can be approximated. First, the covariance of the parameters must be computed with

$$\text{cov}(\theta_{\text{ML}}) \simeq 2 \left(J_{\text{re}}^H J_{\text{re}} \right)^{-1}. \quad (48)$$

The covariance of the poles is then given by

$$\text{cov}(p) = \frac{dp}{d\theta_{\text{ML}}} \text{cov}(\theta_{\text{ML}}) \frac{dp^H}{d\theta_{\text{ML}}}, \quad (49)$$

where $\frac{dp}{d\theta_{\text{ML}}}$ is based on how the parameters $[a_0, \dots, a_{n_a}]$ relate to the poles $[p_0, \dots, p_{n_a}]$.

The resulting standard deviation on the real part of the two poles in the right half-plane was really low (≈ 0.2 rad/s), meaning the estimator evaluates those close to their optimal positions.

For reference, the third pole of the ML estimator, which is visible on the extreme left of Fig. 11 has a standard deviation of ≈ 14500 rad/s. An expected result is the uncertainty on the imaginary port of this pole. Because it is alone, it lays on the real axis and the computed variance is exactly zero. These two clues combined implies that there is definitely a pole on the real axis, but it's exact frequency cannot be extracted from the measured data.

B. Stable ML Estimators

To finally obtain a parametric FRF estimate with a stable model, the model (9) is changed to impose poles in the left half-plane. The transfer function now becomes

$$G(s) = \frac{\sum_{k=0}^{n_b} b_k s^k}{\prod_{k=0}^{n_a} (s + \alpha_k^2)}, \quad (50)$$

which has complex poles as long as the parameters $[\alpha_0, \dots, b_{n_b}]$ are real.